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Recent Trends in Internet-of-Things-Enabled Sensor Technologies for Smart Agriculture

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16-03-2023

***Abstract Abstract*—Smart agriculture integrates key information communication technologies with sensing technologies to provide effective and cost-efficient agricultural services. Smart agriculture leverages a wide range of advanced technologies, such as wireless sensor networks, Internet of Things, robotics, agricultural bots, drones, artificial intelligence, and cloud computing. The adoption of these technologies in smart agriculture enables all stakeholders in the agricultural sector to develop better managerial decisions to get more yield. We differentiate between traditional agriculture and smart agriculture based on the deployment architectures along with a focus on the various processing stages in smart agriculture. We present a comprehensive review of various types of sensors that are playing a vital role in enabling smart agriculture. We also review the integration of various sensing technologies with emerging technologies and computing infrastructures to make agriculture smarter. Finally, we discuss open research challenges that must be addressed to improve the adoption and deployment of smart agriculture in the future.**

Index Terms*—Communication protocols, computing infrastructure, Internet of Things (IoT), sensing technologies, sensors, smart agriculture, smart farming

1 Introduction

AGRICULTURE is considered as one of the most ancient professions of human beings. It provides the major ingredient of food for mankind and other living organisms on earth. By 2050, it is expected that the worldwide population will be 9–10 billion resulting in increased food demand to Manuscript received 28 February 2022; revised 17 August 2022; accepted 21 September 2022. Date of publication 27 September 2022; date of current version 21 November 2022. The work of Sherali Zeadally was supported by the Fulbright U.S. Scholar Grant Award Administered by the U.S. Department of State Bureau of Educational and Cultural Affairs, and through its Cooperating Agency the Institute of International Education (IIE). (Corresponding author: Faisal Karim Shaikh.) Faisal Karim Shaikh is with the Institute of Information and Communication Technologies and the Department of Telecommunication Engineering, Mehran University of Engineering and Technology, Jamshoro 76062, Pakistan (e-mail: faisal.shaikh@faculty.muet.edu.pk). Sarang Karim is with the Institute of Information and Communication Technologies, Mehran University of Engineering and Technology, [h] Jamshoro 76062, Pakistan, and also with the Department of Telecommunication Engineering, Quaid-e-Awam University of Engineering, Science and Technology, Nawabshah 67450, Pakistan (e-mail: sarangkarim@ quest.edu.pk). Sherali Zeadally is with the College of Communication and Information, University of Kentucky, Lexington, KY 40506 USA (e-mail: szeadally@uky.edu). Jamel Nebhen is with the College of Computer Science and Engineering, Prince Sattam Bin Abdulaziz University, AlKharj 11942, Saudi Arabia (e-mail: j.nebhen@psau.edu.sa). Digital Object Identifier 10.1109/JIOT.2022.3210154 60the late 1960s has caused a fundamental change in agricultural systems [3]. To increase the crop yield, farmers started to adapt the use of hybrid seeds, fertilizers, and pesticides rather than relying on conventional agriculture techniques. To optimize the agriculture yield, farmers have to keep track of various factors, such as environmental (temperature, humidity, and CO2) [4], [5], [6], [7], [8], terrestrial (insect detection and leaf chlorophyll) [9], [10], underground (soil temperature, soil humidity, and soil moisture) [2], [11], [12], [13], [14], [15], [16], and irrigation (water flow and level) [12], [17], [18], [19]. Along with open field agriculture, indoor farming is also getting attention because the environment cannot be controlled in the open field. Indoor farming generally referred to as greenhouse farming is a unique mode of growing crops under transparent shelter structures. The main purpose of greenhouses is to provide favorable growing conditions and to protect crops from unfavorable weather and various pests. Greenhouse farming also includes vertical, hydroponic, and aquaponic farming. In vertical farming, the specific crops are grown at various vertical stacked levels compared to only on single plan in an open field. In

Table 1: Comparision of localisation error

Sensing Technology	Reference	Sensing Parameters	Application
Tertill	140	Plant height,weed	Plant
Plant2	139	weed	Animal
Animal	141	Plant height	Plant
Human	143	Plant height	Human
Tertill	121	Plant height,weed	Plant

contrast, hydroponic farming does not require any soil for crop growth and relies on water and nutrients [20]. Aquaponic combines cultivating fish with hydroponics where the aquaculture water is provided to hydroponically grown plants [21]. Recent agricultural advances correspond to the synthetic technological developments in traditional agricultural systems around the world [22]. Although modern agricultural techniques have boosted food production and helped the world to cope with hunger crisis, they have also drastically increased health calamities [23]. For instance, the usage of modern pesticides reduces the antioxidant levels which may lead to neurodegeneration, cardiovascular, respiratory, renal, endocrine, cancer, and reproductive problems [24]. Thus, researchers have started to re-examine and review modern agricultural practices. Recent advances and usage of information and communications technologies (ICTs) in the agricultural sector have brought about numerous benefits and have paved the way toward smart agriculture in both outdoor and indoor environments. ICTs can be used in the various agricultural sectors, such as cultivation, harvesting, irrigation, and farming. ICTs help in improving the production efficiency, production quality, post-harvesting processes, and monitoring of the environmental parameters that affect the agriculture ecosystem. relies on water and nutrients [20]. Aquaponic combines cultivating fish with hydroponics where the aquaculture water is provided to hydroponically grown plants [21]. Recent agricultural advances correspond to the synthetic technological developments in traditional agricultural systems around the world [22]. Although modern agricultural techniques have boosted food production and helped the world to cope with hunger crisis, they have also drastically increased health calamities [23]. For instance, the usage of modern pesticides reduces the antioxidant levels which may lead to neurodegeneration, cardiovascular, respiratory, renal, endocrine, cancer, and reproductive problems [24]. Thus, researchers have started to re-examine and review modern agricultural practices. Recent advances and usage of information and communications technologies (ICTs) in the agricultural sector have brought about numerous benefits and have paved the way toward smart agriculture in both outdoor and indoor environments. ICTs can be used in the various agricultural sectors, such as cultivation, harvesting, irrigation, and farming. ICTs help in improving the production efficiency, production quality, post-harvesting processes, and monitoring of the environmental parameters that affect the agriculture ecosystem.relies on water and nutrients [20]. Aquaponic combines cultivating fish with hydroponics where the aquaculture water is provided to hydroponically grown plants [21]. Recent agricultural advances correspond to the synthetic technological developments in traditional agricultural systems around the world [22]. Although modern agricultural techniques have boosted food

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Figure 1: The structure of our tic-tac-toe implementation.